

CS & IT ENGINEERING

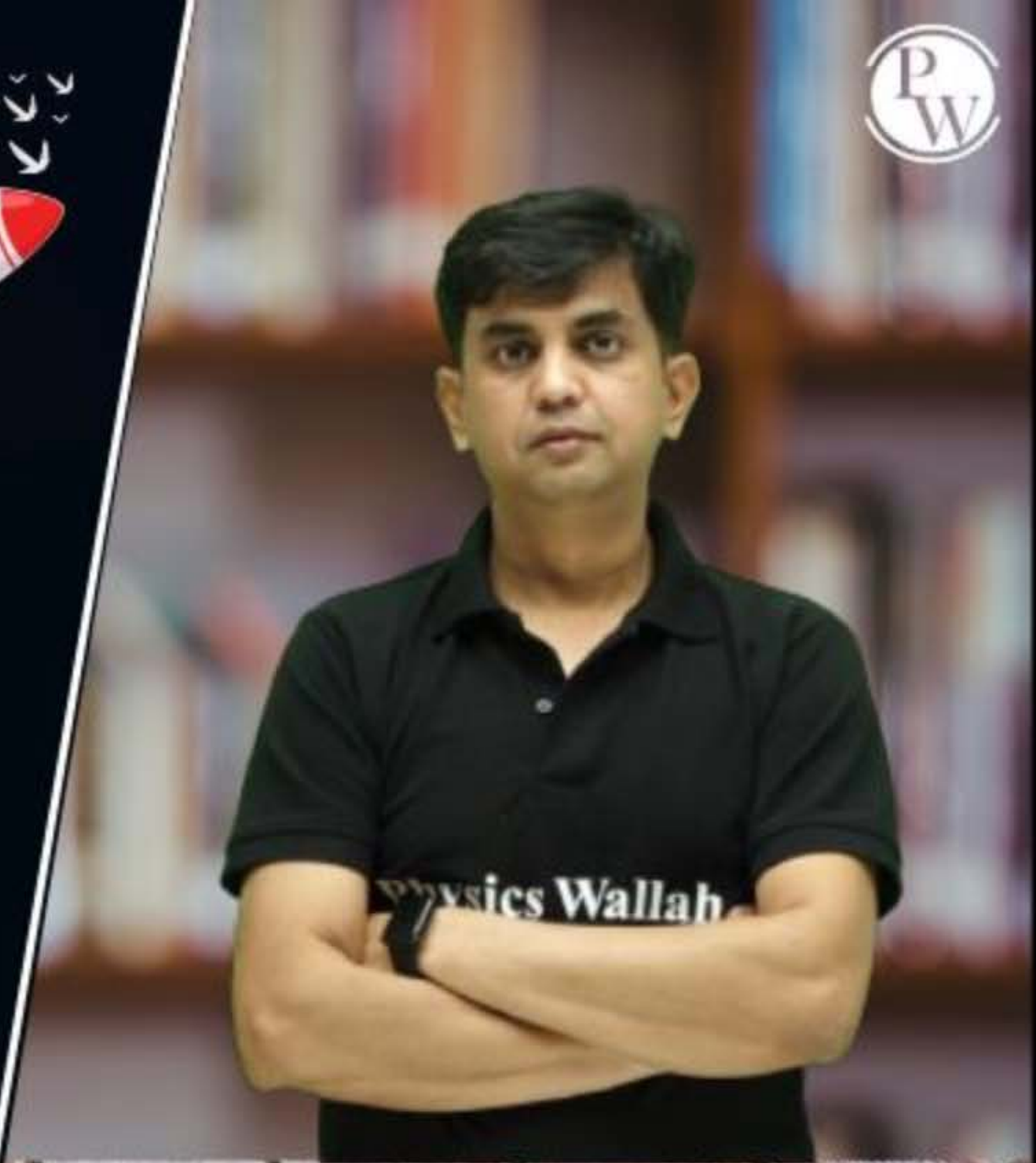


Computer Network

MAC Sub-Layer

Lecture No. - 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

Slotted ALOHA



Topics to be Covered



Topic

CSMA

Topic

CSMA/CD

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support, if each host has to be provided a minimum through put of 0.16 frames per time slot?

[GATE 2004]

~~(A) 1~~

☒ (B) 2

(C) 3

(D) 4

Ans: B

Solution :-

Each host transmits in slot with probability (p) = 0.2

Maximum number of hosts (N) = ?

Throughput of Host = 0.16

Throughput of Host = $p * (1 - p)^{(N-1)}$

$$0.16 = 0.2 * (0.8)^{(N-1)}$$

$$\text{for } [N=2]$$

#Q. Consider a LAN with four nodes S_1 , S_2 , S_3 and S_4 . Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot by S_1 , S_2 , S_3 and S_4 are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.

[GATE 2015]

Solution :-

Node S1 transmits in a slot with probablity (P_1) = 0.1

Node S2 transmits in a slot with probablity (P_2) = 0.2

Node S3 transmits in a slot with probablity (P_3) = 0.3

Node S4 transmits in a slot with probablity (P_4) = 0.4

Throughput of Node S1 = $(P_1) * (1 - P_2) * (1 - P_3) * (1 - P_4)$

Throughput of Node S2 = $(1 - P_1) * (P_2) * (1 - P_3) * (1 - P_4)$

Throughput of Node S3 = $(1 - P_1) * (1 - P_2) * (P_3) * (1 - P_4)$

Throughput of Node S4 = $(1 - P_1) * (1 - P_2) * (1 - P_3) * (P_4)$

$$\text{Throughput of Channel} = \left[\begin{aligned} &\text{Throughput of Node S1} \\ &+ \text{Throughput of Node S2} \\ &+ \text{Throughput of Node S3} \\ &+ \text{Throughput of Node S4} \end{aligned} \right]$$

$$\begin{aligned} \text{Throughput of Channel} &= (0.1 * 0.8 * 0.7 * 0.6) \\ &+ (0.9 * 0.2 * 0.7 * 0.6) \\ &+ (0.9 * 0.8 * 0.3 * 0.6) \\ &+ (0.9 * 0.8 * 0.7 * 0.4) \\ &= 0.4404 \end{aligned}$$



Topic : CSMA

⇒ Multiple Access Protocol
(Random Access)



- Carrier Sense Multiple Access
↓
- Sense before transmit
[Sense the channel, before transmission]
- if channel sensed idle : "transmit entire frame"
- if channel sensed busy : "defer transmission"



Topic : CSMA



Different variations of CSMA protocols :

- i. 1 – Persistent CSMA
- ii. Non – Persistent CSMA
- iii. p – Persistent CSMA



Topic : 1-persistent CSMA



→ Sense before transmit

→ if channel sensed idle

transmit entire frame immediately (with probability 1)

→ if channel sensed busy

sense the channel continuously (till it become idle)

⇒ Better to use in case n/w have low traffic



Topic : Non-persistent CSMA



- Sense before transmit
- if channel sensed idle
transmit entire frame immediately
- if channel sensed busy
Wait a random amount of time and then sense the channel

⇒ Better to use
in case of high
traffic

It do not sense the channel continuously when channel is busy.
[Unlike 1 - Persistent CSMA]



Topic : p-persistent CSMA



→ Channel has divided into time slots

→ Sense before transmit

→ if channel sensed idle

transmit the frame with probability p in that idle slot

→ if channel sensed busy

sense the channel continuously (till it become idle)

$$0 < P \leq 1$$

For high traffic
 $P \rightarrow 0$

For low traffic
 $P \rightarrow 1$

⇒ CSMA leads lesser no. of collisions in compare to ALOHA (due to sense before transmit).



Topic : CSMA/CD



- CSMA with Collision Detection
- Applicable only for wired LAN (Bus topology)
- Sense before transmit ✓
- Sense while (during) transmission ←
- No any feedback (acknowledgment) from receiver



Topic : CSMA/CD



→ To detect collision,
minimum frame transmission delay should be
greater than equal to (maximum) round trip propagation delay.

frame transmission delay $\geq$ round trip propagation delay

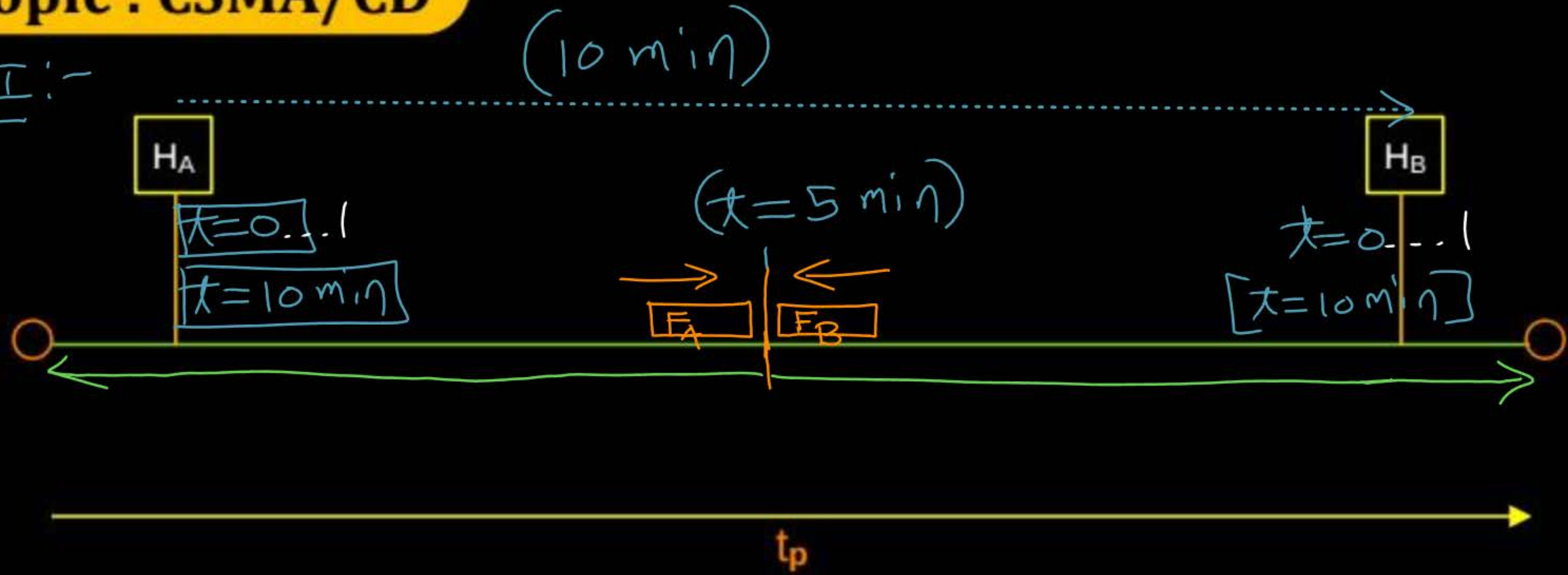
$$t_x \geq 2 * t_p$$



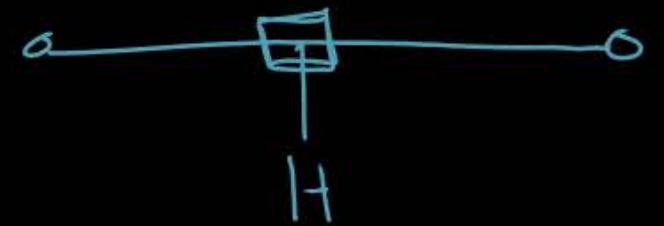
Topic : CSMA/CD



CASE I:-



[one frame transmission time = 1 min]

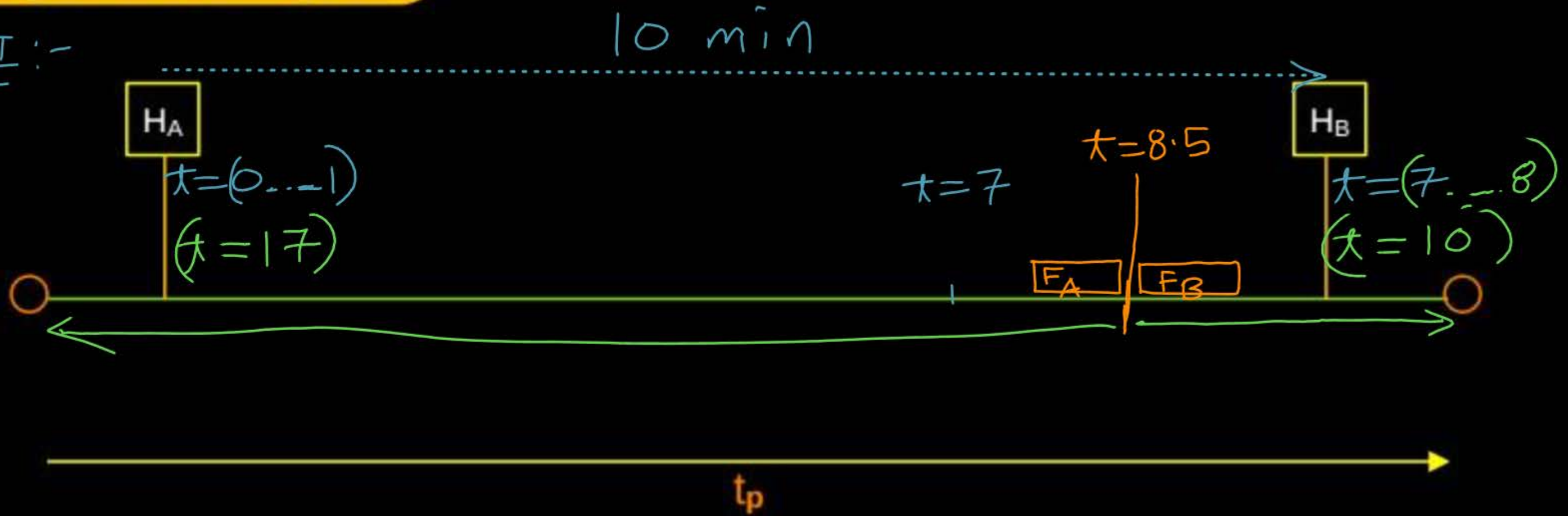




Topic : CSMA/CD



CASE II :-



one frame transmission time = 1 min



Topic : CSMA/CD



CASE III

10 min





2 mins Summary



Topic

CSMA ✓

Topic

CSMA/CD



THANK - YOU

